AI-Driven Fabric Defect Detection & Monitoring System for the Appeal Industry in Sri Lanka

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***Abstract*— This paper introduces FabricVision, an AI-driven quality monitoring system designed to enhance fabric inspection processes in Sri Lanka's apparel industry. By automating the traditional manual and inconsistent inspection of fabric rolls, the system supports pre-seam fabric quality assessment through the integration of high-resolution imaging, motion-aware frame selection, edge-level fog processing and advanced deep learning models. Intelligent fabric capture combined with anomaly pre-screening and lightweight quality analytics ensures that only reliable and meaningful fabric data is processed, improving inspection efficiency. An edge-level fog processing layer further supports this pipeline by performing adaptive image enhancement based on fabric appearance characteristics, improving visual consistency before defect analysis. You Only Look Once is employed for real-time detection and localization of defects such as holes, stains and fabric irregularities, enabling meter-level mapping of flaws on fabric rolls. Beyond defect detection, the system improves inspection consistency, reduce latency, and support proactive quality control through predictive and prescriptive analytics that integrate detection results with temporal trend analysis. In doing so, the system delivers a practical and scalable solution for fabric quality monitoring in resource-constrained apparel manufacturers.**

Keywords*— *Intelligent frame capture, fog computing, real-time defect detection, predictive Analytics

I. Introduction

The apparel manufacturing industry is a major contributor to Sri Lanka's economy, generating substantial employment and export revenue [1]. As production volumes increase and global quality expectations become more stringent, manufacturers face growing challenges in maintaining consistent fabric quality under high-speed industrial conditions. Effective and reliable inspection mechanisms are therefore essential to reduce defects, minimize waste and sustain competitiveness. In this context, fabric quality assessment during the pre-seam phase where fabric rolls are inspected before being selected, cut or fed into production lines plays a critical role in ensuring downstream production reliability.

Undetected fabric defects at this early stage can propagate through subsequent processes, leading to material wastage, rework and inconsistent product quality [2]. Despite its importance, fabric roll inspection in many small-scale apparel factories remains heavily dependent on manual visual examination. In small and medium-scale apparel manufacturing environments, fabric inspection is typically performed by human operators without the assistance of specialized inspection machinery. Such manual processes are highly subjective and depend on individual experience, visual acuity and environmental conditions [3].

This research introduces FabricVision, a modular real-time fabric inspection system designed to support pre-seam fabric quality assessment and informed fabric roll selection. The end-to-end workflow integrates intelligent fabric capture with anomaly pre-screening and quality analytics, edge-level fog processing and image enhancement, real-time defect detection and localization, and predictive analytics for forecasting fabric quality trends. Together, these modules enable early quality assurance and proactive decision-making before seam formation.

The proposed system performs intelligent fabric capture and pre-screening at the pre-seam stage using an edge-based imaging setup consisting of a Raspberry Pi Camera Module 3 and a Raspberry Pi 4 Model B. The module is designed to selectively acquire fabric frames during active fabric movement, avoiding unnecessary data capture during idle states. Captured frames are preliminarily assessed through anomaly awareness and lightweight quality analysis to ensure that meaningful and reliable fabric data proceeds to subsequent inspection stages.

Within the proposed system, a fog-level image enhancement and analysis stage is introduced at the edge layer to improve fabric image quality prior to defect detection. The system applies lightweight fabric pattern classification and adaptive enhancement techniques to stabilize visual quality under varying lighting and texture conditions. Real-time quality metrics, including brightness, contrast, and sharpness, are computed to guide enhancement decisions and to provide interpretable indicators for downstream inspection stages. These fog-level operations ensure consistent fabric representations while reducing computational overhead for later inspection stages.

For real-time defect detection and localization, the system combines an optimized You Only Look Once version 8 (YOLOv8) model with rotary encoder tracking. This system identifies defects like holes, stains, and cuts and maps each to an exact meter position on the fabric roll. Operators receive instant visual and numerical feedback, allowing quick corrections. This reduces manual inspection, cuts fabric waste, lowers rework, and improves quality and efficiency.

Building on the localized defect information, a predictive and prescriptive analytics module uses historical and real-time data to forecast future defect locations along the roll. It performs risk scoring and root-cause analysis identifying issues like material or supplier faults and provides actionable insights for early intervention and data-driven quality control, all without impacting real-time inspection performance.

II. Related Work

Maintaining consistent fabric quality remains a long-standing challenge in the apparel manufacturing industry. Traditional fabric inspection processes rely heavily on manual visual assessment, which is labor-intensive, time-consuming, and inherently subjective [4]. Variations in operator experience, environmental conditions, and inspection fatigue often result in inconsistent quality judgments and missed defects. These limitations have been widely reported to contribute to increased material waste, rework, and production inefficiencies, particularly in small and medium-scale garment manufacturing environments where advanced inspection machinery is not readily available.

To overcome these challenges, numerous computer vision-based fabric inspection systems have been proposed. Early approaches focused on handcrafted texture features and statistical analysis for defect identification [4]. However, these systems are predominantly designed for in-line or post-seam inspection, assuming continues, high-quality video input and predefined defect patterns. As a result, they often overlook issues related to redundant data capture, frames quality validation, variability, and the presence of previously unseen fabric irregularities.

Recent research has highlighted the importance of edge-level processing and intelligent data reduction to support real-time inspection in resource-constrained environments [5]. Edge-based systems utilizing low-cost cameras and embedded devices such as Raspberry Pi platforms have shown potential for reducing latency and minimizing dependence on cloud infrastructure. Nevertheless, most existing approaches focus primarily on defect detection, providing limited support for upstream pre-seam inspection, including motion-aware frame selection, unsupervised anomaly identification, and frame-level quality validation prior to defect detection [6].

Unsupervised anomaly detection and image quality assessment techniques have also been widely investigated in computer vision research, particularly for general surface inspection and texture analysis tasks [7]. Autoencoder-based and reconstruction-driven approaches have shown potential in identifying abnormal patterns without requiring labeled defect data, while frame-level quality metrics have been used to evaluate blur, exposure imbalance, noise, and texture consistency [8]. Within fabric inspection systems, anomaly detection and quality assessment are typically applied as separate or offline processes, with limited research integrating them as unified, real-time pre-inspection stage operating before defect detection in resource-constrained environments.

Industrial visual inspection systems are highly dependent on image quality for reliable surface analysis and defect detection. In real manufacturing environments, fabric images are often degraded by uneven illumination, noise, and low contrast, motivating the use of image enhancement and edge information preservation techniques. Prior studies have shown that enhancing edge information improves inspection reliability under challenging conditions [9], while computationally efficient edge enhancement methods can enhance sharpness and support real-time processing requirements [10].

Recent studies highlight the importance of preserving local texture and edge continuity for reliable industrial inspection, showing that selective enhancement is essential to balance noise suppression and structural preservation in textured surfaces such as fabrics [11]**.** However, most existing approaches treat image enhancement as a standalone preprocessing step, lacking semantic awareness of surface patterns and real-time quality assessment, which motivates the proposed fog-level, pattern-aware adaptive enhancement strategy.

Traditional manual fabric inspection is slow and error-prone on modern production lines [12]. Deep learning, especially Convolutional Neural Networks (CNN) and YOLO algorithms, has greatly improved accuracy and speed in detecting defects like holes and stains [13]. However, a major limitation persists, these systems typically only identify defects within the camera image and cannot locate them precisely on a continuously moving fabric roll. This exact physical localization is essential to reduce waste and improve efficiency.

While real-time defect detection has improved, locating defects on moving fabric is still a challenge. Existing methods like physical markers disrupt production. Although IoT sensors like rotary encoders are widely used in automation, their integration with deep learning vision systems for textile inspection is underexplored [14]. This work addresses the gap by integrating YOLOv8-based defect detection with rotary encoder tracking, enabling precise defect localization without disrupting production.

Existing inspection systems are reactive, detecting defects in real time but unable to predict future ones along a fabric roll [15]. Predictive analytics [16], using sequence-based models like Long Short-Term Memory (LSTM) for defect forecasting, and ensemble models like XGBoost and Random Forest for roll-level risk scoring and root-cause analysis [17], can address this. Explainable AI tools like SHAP enhance the trustworthiness of these AI-driven quality systems.

III. Methodology

The Overall system architecture, illustrated in Fig.1, visualizes the complete system procedure with all four components together.

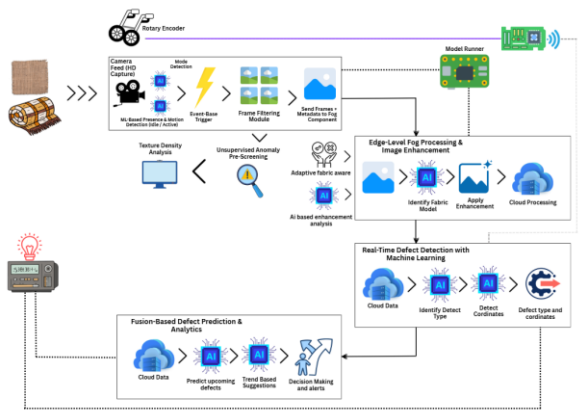

Fig. 1. Overall System Architecture Diagram

A. *Intelligent Frame Capture and Pre-Inspection*

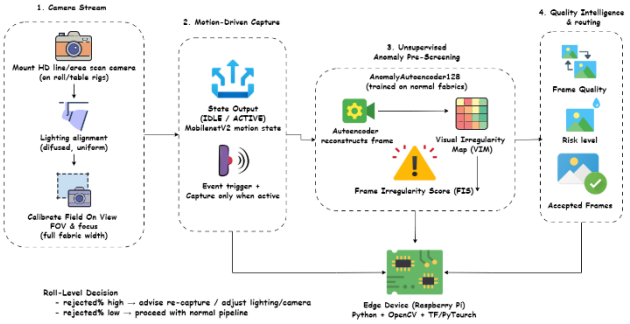

Fig. 2. Motion base Frame Extraction and Pre-Inspection Diagram

As shown in Fig.2, the Intelligent Frame Capture and Pre-Inspection component operates at the pre-seam fabric roll level and serves as the initial stage of the FabricVision system.

1) Data Acquisition and Edge Hardware Setup: Fabric surface images are acquired using a Raspberry Pi Camera Module 3 connected to a Raspberry Pi 4 Model B, forming a compact edge-based imaging unit. The camera is positioned to capture the surface of fabric rolls during pre-seam inspection, allowing visual assessment of fabric texture and appearance before cutting or production use. The imaging system captures high-resolution frames under controlled lighting conditions to ensure consistent visual quality across different fabric types. Data acquisition is performed locally on edge device, enabling real-time processing without reliance on continuous cloud connectivity.

2) Motion-Aware Frame Selection: Continuous frame capture during fabric inspection can generate large volumes of redundant or uninformative data, particularly when the fabric is stationary or moving irregularly. To address this, the proposed system incorporates a motion-aware frame selection mechanism that synchronizes frame acquisition with encoder pulse measurements obtained from the conveyor system and monitors temporal changes in the image stream to differentiate between active and idle fabric states, capturing frames only during meaningful fabric movement. This approach reduces unnecessary data generation and computational overhead, improves processing efficiency.

3) Anomaly Pre-Screening: After motion-aware frame selection, captured frames undergo anomaly pre-screening to identify abnormal texture behavior prior to defect detection. This unsupervised process models the normal fabric appearance to detect subtle or previously unseen irregularities and generates a frame-level abnormality indicator. Abnormal frames are prioritized for further analysis, while normal frames proceed through the pipeline, improving inspection robustness and reducing the influence of unreliable data on downstream processing.

4) Frame-Level Quality Analytics: In addition to anomaly awareness, each captured fabric frame is assessed using frame-level quality analytics to evaluate its visual reliability prior to inspection. Lightweight measures quantify key attributes such as sharpness, exposure balance, noise, and texture consistency in real time on edge devices, enabling early detection of degraded frames. These quality indicators guide inspection decisions by filtering low-quality frames, and persistent quality degradation across a fabric roll triggers roll-level alerts to support operator actions, thereby improving inspection consistency and downstream defect detection reliability.

B. *Edge-Level Fog Computing Image Enhancement & Quality Analysis*

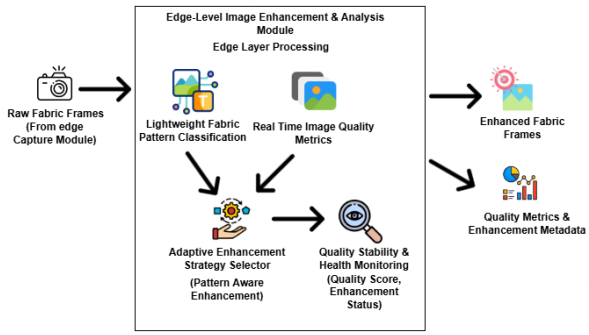

Fig. 3. Edge-Level Image Enhancement & Analysis Diagram

As shown in Fig. 3, this edge-layer component integrates a fog-level image enhancement and quality analysis stage to processes raw fabric frames in real time, stabilizing visual quality under varying illumination while reducing latency and computational overhead for downstream inspection modules.

1) Lightweight Fabric Classification: To enable fabric-aware image enhancement, a lightweight fabric classification module is deployed at the fog layer. Incoming frames are first categorized as patterned or non-patterned fabrics using an EfficientNet-Lite convolutional neural network, selected for its low computational overhead and suitability for edge devices. Frames identified as patterned fabrics are further classified into four representative pattern categories: stripe, check, floral, and geometric patterns. This secondary classification is performed using an EfficientNet-B0 model, which provides improved feature extraction capacity for distinguishing fine texture variations between pattern types.

2) Real-Time Image Quality Metric Computation: In parallel with pattern classification, real-time image quality metrics are computed to assess visual conditions. Brightness is estimated using mean pixel intensity, contrast is derived from grayscale intensity dispersion, and sharpness is measured using the variance of the Laplacian operator. These computationally efficient metrics are suitable for fog-level deployment and have been shown to support edge and texture preservation under uneven illumination [9].

3) Adaptive Enhancement Strategy Selection: Based on the outputs of pattern classification and image quality metrics, the system selects an adaptive enhancement strategy to stabilize visual quality while preserving fabric texture details. The enhancement pipeline begins with automatic white balancing using the gray-world assumption, followed by gamma correction to normalize brightness. Contrast

Limited Adaptive Histogram Equalization is then applied to improve local contrast and highlight seam structures. For patterned fabrics, edge-preserving filtering is applied to suppress noise while maintaining pattern edges and textures. In contrast, non-patterned fabrics are processed using bilateral filtering to achieve smoother noise reduction in uniform regions. Finally, unsharp masking is used to enhance seam boundaries and structural edges important for defect detection. Enhancement parameters are dynamically adjusted based on quality metrics and pattern context to prevent over-processing artifacts.

4) Output Integration with Downstream Inspection: The fog-level processing stage outputs enhanced fabric frames and associated quality metadata. Enhanced frames are forwarded to downstream defect detection modules, while quality metrics and enhancement status information are provided to the system's monitoring and visualization layer, enabling efficient integration and transparent enhancement behavior.

C. Fabric Defect Detection and Localization using YOLO and IoT-Based Rotary Encoder Tracking

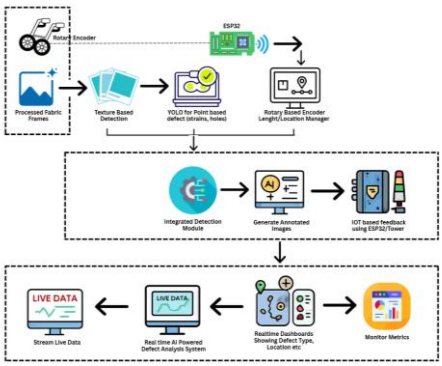

Fig. 4. Defect Detection & Localization using IoT Diagram

As illustrated in Fig. 4, the FabricVision system combines YOLO-based computer vision defect detection with IoT-enabled physical position tracking, overcoming the limitations of vision-only systems by enabling precise, real-time localization of defects on moving fabric.

1) Defect Detection with YOLOv8: YOLOv8 is used for efficient feature extraction and high-accuracy real-time defect detection, which are critical for inspecting continuously moving fabric in industrial environments. The model processes captured frames to identify defects like holes, stains, and cut irregularities. During detection, YOLOv8 analyzes each incoming frame, outputting defect classifications alongside bounding box coordinates. YOLOv8 advanced architecture learns spatial feature hierarchies, enabling accurate distinction between visually similar defect types across different textures and lighting conditions. During the training process, YOLOv8 demonstrated strong performance, achieving high precision and recall across multiple defect categories, confirming its reliability for real-time fabric inspection.

2) Encoder Assisted Fabric Position Tracking: To accurately localize fabric defects, the system uses an IoT-based, encoder-assisted hardware tracking mechanism integrated with defect detection. A rotary encoder with a precision measuring wheel is mechanically attached to the fabric feed mechanism to measure linear fabric movement. The encoder generates pulse signals proportional to fabric displacement, which are processed in real time by an ESP32 Nano microcontroller. The encoder's pulses are converted into precise linear distance measurements. Each measurement is time-stamped and synchronized with a camera frame, ensuring every detected defect is mapped to an exact position along the fabric roll.

3) Data Fusion and Localization Engine: To find the exact position of each defect on the fabric, the system combines the results from the camera and rotary encoder. When YOLOv8 detects a defect in an image, the system takes the position of that defect in the frame. At the same time, the rotary encoder provides the exact length of fabric that had moved through the rotary encoder-based device. The data fusion engine combines camera and fabric movement data to convert a defect image position into its precise location in meters along the fabric roll. This allows operators to instantly find defects, cutting waste and boosting inspection accuracy.

4) Realtime Operator Feedback and Visualization Framework: When a defect is detected, the system provides immediate operator feedback. An IoT interface (ESP32) activates color-coded visual alerts and displays the defect type and exact location in meters. Annotated images are saved for records. The system integrates YOLOv8 detection with rotary encoder tracking for accurate localization on moving fabric. This synchronized approach improves inspection efficiency, reduces waste, and supports data-driven quality assurance.

D. Real-time future fabric defect prediction & prescriptive analytics for risk assessment

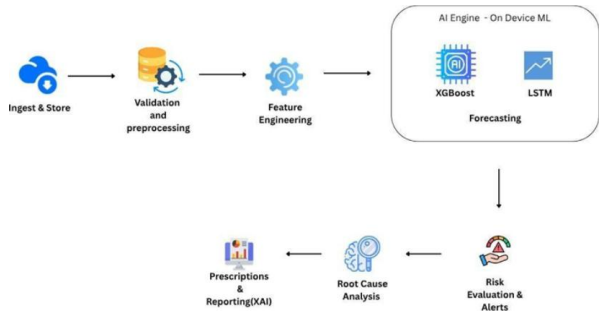

Fig. 5. Future defect prediction & prescriptive analytics diagram

The Predictive and Prescriptive Analytics component of FabricVision extends real-time defect detection by predicting future defects and assessing roll-level risk. It uses a sequence-based prediction model and a roll-level risk and root cause analysis model to convert defect data into actionable insights for operators and supervisors.

1) Feature Engineering and Data Aggregation: Defect level data captured during fabric inspection including defect positions, timestamps, severity labels, and roll metadata are first preprocessed and aggregated at the fabric roll level. For each roll, statistical and spatial features are computed to characterize defect behavior over the inspected length. These features include defect count, defect density, average defect

severity, mean inter defect distance, standard deviation of inter-defect intervals, roll length, and supplier identifiers encoded numerically. This aggregation process converts high frequency defect events into structured representations suitable for downstream predictive modeling, while preserving key information about defect intensity, dispersion, and temporal consistency.

2) Sequence-Based Future Defect Prediction (Model A): This model is designed to predict the spatial progression of defects along the fabric roll by learning temporal patterns from recent defect sequences. The model operates on fixed length sequences comprising the last ten observed defect positions or inter defect intervals. By analyzing these sequences, the model captures trends such as regular spacing, drift, or sudden clustering of defects. The trained sequence-based model outputs the predicted locations of the next ten potential defects along the fabric roll. This forward prediction enables early inspection planning, targeted quality checks, and preemptive intervention before defects propagate further along the production line.

3) Roll Level Risk and Root Cause Analytics (Model B): This model provides higher-level analytical insight using three machine learning models. An XGBoost regression model estimates a continuous roll level risk score based on aggregated defect features. A Random Forest classifier categorizes defect distributions into repeating, drifting, or irregular patterns, while a second Random Forest model identifies the most likely root cause (material, supplier related). These models are trained on historical roll data and validated on unseen samples to ensure robust performance.

4) Explainability, Prescriptive Insight, and Dashboard Integration: To improve transparency, SHAP-based explainability is applied to the risk model, highlighting feature contributions behind each prediction. Based on risk scores, pattern classes, and root cause outputs, the system generates prescriptive insights such as recommended inspection zones or targeted corrective actions. All outputs are integrated into an operator-facing dashboard.

IV. Results and Discussions

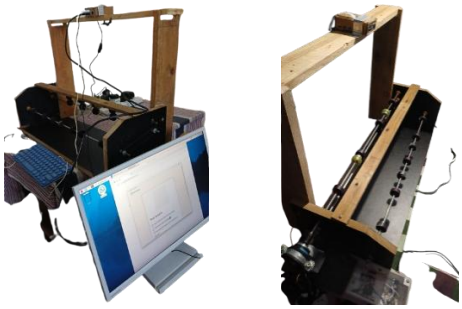

Fig. 6. Prototype of the FabricVision system developed.

In the experimental evaluation of the proposed edge-level fabric capture and pre-screening module the motion-driven capture pipeline enabled spatially consistent frame acquisition along the fabric roll through encoder pulse synchronization, ensuring reliable sampling of the moving fabric surface. The anomaly pre-screening stage successfully identified irregular fabric regions through the computed Frame Irregularity Score (FIS), enabling the system to categorize frames into normal, warning, and anomalous groups during real-time operation. In addition, the frame-level quality analytics stage effectively filtered visually unusable frames affected by blur or illumination inconsistencies using the computed Frame Quality Score (FQS). The complete pipeline maintained real-time processing performance of approximately 28–30 FPS while filtering about 4–6% of frames that did not meet the quality threshold.

The fabric-aware preprocessing module was evaluated to measure its impact on adaptive image enhancement prior to defect detection. The pattern classification module achieved an accuracy of approximately 92-94%, enabling reliable identification of fabric textures and pattern structures. Using these classification outputs, adaptive enhancement techniques improved brightness, contrast, and texture visibility in captured frames. The preprocessing pipeline maintained real-time performance on the edge device, achieving an average processing speed of 22-25 FPS, while contributing to an overall 4-6% improvement in downstream defect detection accuracy.

The YOLOv8-based defect detection module demonstrated strong performance during experimental evaluation, achieving a mean Average Precision (mAP@50) of 94.7%, indicating high reliability in detecting fabric defects such as holes, stains, and irregular cuts across varying fabric textures and lighting conditions. When integrated with the rotary encoder-based localization mechanism, the system was able to determine the physical position of detected defects on the fabric roll with an accuracy of approximately ±2 cm. This precise localization capability allows operators to quickly identify and isolate defective regions during inspection. Compared with traditional manual inspection processes, which typically achieve only 75–85% accuracy and are prone to human fatigue and delayed detection, FabricVision provides faster and more consistent defect identification along with exact meter-level location mapping. The combination of real-time detection and IoT-based position tracking therefore significantly improves inspection efficiency and supports more reliable quality control decisions in pre-seam inspection environments.

Strong performance was shown by the Predictive and Prescriptive Analytics component in identifying defect patterns and producing quality insights for the future based on inspection data from the past. The sequence-based defect forecasting model achieved an R^2 accuracy of approximately 96%, effectively capturing temporal patterns and predicting the next 30 potential defect locations along the fabric roll. Additionally, the roll-level analytics module showed reliable performance, with the XGBoost-based risk scoring model achieving 94.5% accuracy and the pattern identification and root-cause classification model achieving 95.1% accuracy. These models successfully categorized defect distributions into repeating, drifting, and irregular patterns and identified possible material-related causes. Overall, the framework provides interpretable risk assessments and early warnings, enabling proactive quality management and helping reduce fabric waste in textile manufacturing environments Quantitative Evaluation of Defect Detection.

TABLE 1: DEFECT DETECTION PERFORMANCE

Defect Class	Precision	Recall	F1 Score	mAP@50
Hole	0.94	0.93	0.935	0.95
Stain	0.93	0.92	0.925	0.94
Cut	0.92	0.90	0.910	0.93
Lines	0.88	0.85	0.865	0.87
Overall	0.9288	0.9126	0.9206	0.9475

Table 1 shows the YOLOv8 model achieved 92.88% precision, 91.26% recall, 92.06% F1-score, and 94.75% mAP@50, with hole defects detected most accurately and line defects being more challenging. Overall, the results confirm reliable real-time detection performance for fabric inspection.

A. *Strengths of FabricVision*

The experimental evaluation highlights several strengths of the proposed system. Its pre-seam inspection capability enables early identification of fabric quality issues before cutting or production, reducing material waste and rework. By integrating intelligent frame capture, anomaly awareness, and quality analytics at the edge, the system ensures that only reliable and meaningful fabric data is processed, while edge-level fog processing supports low-latency operation and reduces dependence on continuous cloud connectivity.

The defect detection component enables accurate real-time identification and meter-level localization of fabric defects by combining YOLOv8-based vision with synchronized IoT rotary encoder tracking. Detected defects are visually highlighted with precise physical coordinates, allowing operators to quickly assess defect location and severity. In addition, the predictive and prescriptive analytics component analyzes historical and streaming data to identify quality trends, supporting proactive decision-making and early intervention before quality degradation escalates.

B. *Limitations of the system*

1) Hardware Scalability: Inspection of long fabric rolls may affect system responsiveness on resource-constrained edge devices due to increased processing duration and data volume. While the system was evaluated on a Raspberry Pi 4 Model B (8 GB RAM, 64 GB storage) and Raspberry Pi Camera Module 3 is suitable for small-scale settings, higher inspection accuracy may require industrial-grade cameras under demanding conditions and also stable lighting.

2) Performance with paterned fabric: Defect detection model often struggle with patterned fabrics. Their complex designs and repetitive textures can cause false positives (patterns mistaken for defects) and false negatives (real defects missed).

V. CONCLUSION

This paper presented FabricVision, a modular AI-driven fabric inspection system for pre-seam fabric quality assessment in small and medium-scale garment manufacturing environments. By integrating intelligent frame capture, edge-level fog processing, real-time defect detection, and predictive analytics, the system enables early identification of fabric quality issues before cutting or production. Experimental results show reduced redundant data capture, improved visual consistency, and accurate defect localization with low latency on resource-constrained edge hardware. Despite dependence on imaging conditions and hardware capabilities, the proposed framework offers a practical and scalable foundation for improving fabric quality control and reducing material waste in the apparel industry.

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